Rajat Moona receiving degree from IIT Kanpur

The birth of a teacher

At IISc, Rajat was selected for two programmes—Electrical Engineering and Computer Science. Choosing the latter, Rajat converted his MSc Engineering programme into a doctorate (PhD) programme after a year. He worked with the most renowned professors of that time, including Prof. V. Rajaraman, on his thesis on building a supercomputer—a parallel machine.

Prof. Rajaraman is known for his extensive work in programming and computer architecture, advisory roles in government programmes, and has published many books, including one on the history of computing in India.

"Many of the major figures in Indian computing are either his students or associated with him. He was a key figure behind the introduction of the MCA (Master of Computer Applications) programme, which expanded India's computing workforce. His books on computing and programming made education accessible and affordable in India. I consider myself fortunate to have done my PhD under his guidance," narrates Rajat, all along his eyes dazzling with awe for his *guru*.

Working under Prof. Rajaraman left an indelible imprint on Rajat who had developed an innate desire to contribute to his country and follow in the footsteps of his teacher. He completed his PhD in 1989 but continued working as a Scientific Officer on a project that involved porting the AT&T SVR 3.2 Unix OS on the machine he had developed, with constructions to do parallel computation. The supercomputer proved very useful once fully functional. "I spent another year at IISc, guiding several students working on their master's theses using my machine. It gave me great satisfaction to know that some of them graduated from IISc after working on their projects with my help," he says with a content smile resting on his face.

Soon after he received an offer from IIT Kanpur and continued his academic career at IIT Kanpur.

After submitting his thesis, Rajat married Rajni, a graduate from IIT Delhi in Mathematics. "One of the reasons I chose to work in Bengaluru for a year was so that both of us could live together and experience the city. After I joined IIT Kanpur, my wife has played a very supportive role in all my work and endeavours, always walking alongside me. When jobs were limited, she taught in schools in Kanpur," Rajat shares.

Currently she works in IIT Bombay in administration in the Department of Electrical Engineering. The couple has a son who followed in his father's footsteps, graduating from IIT Kanpur. With a dual degree (BTech and MTech), he chose the road not taken by his father and is currently working to build industrial advancements for Meta (formerly Facebook) in London.

A teacher, a researcher, a maker

Since 1991, Rajat has been a professor, teaching, conducting research, and guiding students. In 1994, he served as a visiting researcher at the Massachusetts Institute of Technology in the Lab for Computer Science (CSG Group), collaborating with Prof. Arvind's Research Group under a DST fellowship. Around this time, he was approached by the National Informatics Centre (NIC), part of the Ministry of Electronics and Information Technology (then known as the Ministry of Information Technology).

"They were facing an issue related to smart cards for driving licences and asked if I could assist. I examined the problem, which involved creating a smart card-based solution for driving licences.